

PRODUCTVISION

xOS

The Experimentation Operating System

Replace 80% of manual CRO work. Scale experimentation from chaos to system.
From strategic goals to test results to executive decisions — in one intelligent brain.

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CONFIDENTIAL

1. Executive Summary

xOS is an AI-powered experimentation operating system that fundamentally changes how companies run growth experiments. Built on Claude Code with specialized skills and MCP integrations, xOS acts as an intelligent brain that manages the entire experimentation lifecycle: from connecting test ideas to company strategy, through experiment design and execution monitoring, to generating executive-ready reports with clear next-step recommendations.

The core value proposition is radical: companies currently employing 8–12 CRO specialists can achieve the same (or better) output with just 2 human operators working alongside xOS. The system doesn't just automate tasks — it thinks strategically, learns from every experiment, and compounds knowledge over time.

THE OPPORTUNITY

The global A/B testing market is projected to reach \$1.6B by 2027. Yet most companies still run experimentation as a fragmented, manual process. xOS bridges this gap by turning experimentation from a craft into a scalable operating system. The first company to crack this owns the category.

2. The Problem We're Solving

Experimentation at most companies is broken in five fundamental ways:

2.1 Fragmentation

Ideas live in Slack. Hypotheses are in Google Docs. Experiments run in VWO or Optimizely. Results get pasted into PowerPoint. Learnings disappear into email threads. There is no single system of record, and knowledge leaks at every handoff.

2.2 No Strategic Connection

Most experiments are disconnected from company goals. Teams run tests based on gut feeling or HiPPO (Highest Paid Person's Opinion) rather than a structured link from OKRs/OPRs down through KPI trees and opportunity solution trees to specific hypotheses. This means even winning experiments don't always move the metrics that matter.

2.3 Slow Decision Cycles

After an experiment concludes, it takes 1–3 weeks to analyze results, build a report, get it reviewed, and reach a decision. By then, momentum is lost. Management doesn't trust the data because the reports are inconsistent in format and depth. Critical decisions stall.

2.4 Knowledge Doesn't Compound

When Test #47 finishes, the learnings rarely inform Tests #48–#55. There's no systematic way to feed findings back into the hypothesis pipeline. Companies re-learn the same lessons quarter after quarter.

2.5 Expensive and Unscalable

A mature experimentation program requires hypothesis writers, test designers, QA engineers, data analysts, and program managers. For enterprise companies, this means 8–12 full-time specialists at a loaded cost of €800K–€2M per year. Even then, velocity is limited to 3–5 tests per month.

3. The xOS Vision

xOS is an experimentation brain that lives inside Claude Code. It's not a tool — it's an operating system. It handles the thinking, the execution, and the learning. A human stays in the loop for strategic judgment and approval, but the heavy lifting — the 80% of CRO work that is process, analysis, and documentation — is handled by xOS.

The Two-Layer Architecture

Inspired by the conversation with Mateusz from the Second Brain project, xOS separates into two distinct layers:

Layer	What It Contains	Why It Matters
Operating Engine	Decision frameworks, analysis methodologies, prioritization models, strategic alignment logic, reporting templates. This is the secret sauce.	This is what you sell. It's Bjarn's 15+ years of experimentation expertise encoded into skills and agents. It's NOT open-sourced.
Context Layer	Company OKRs, KPI definitions, brand guidelines, audience segments, historical experiment data, tool credentials.	This is customer-specific. Each client feeds their own context. The operating engine processes this context to produce intelligent outputs.

KEY INSIGHT FROM MATEUSZ

Package the operating engine behind MCPs so clients get the outputs but not the secret sauce. They can't see how the sausage is made — they just get brilliant experimentation on autopilot. This is critical for productization and protecting your IP.

The Human-in-the-Loop Model

xOS doesn't eliminate humans. It eliminates the 80% of manual, repetitive, process-driven work. The remaining 20% — strategic judgment, creative ideation, stakeholder relationships, and final approval — requires 2 skilled operators instead of 10. These operators become "Experimentation Pilots": people who fly the xOS system rather than doing the grunt work themselves.

4. Core Modules

xOS consists of nine interconnected modules, each built as a Claude Code skill or agent. Together, they form a complete experimentation lifecycle.

#	Module	Function	Key Integrations
1	Strategic Alignment Engine	Connects every experiment to company OKRs, KPI trees, and opportunity solution trees	Airtable, Google Sheets, Notion
2	Hypothesis Factory	Transforms raw ideas into structured, prioritized hypotheses	Airtable, Miro, internal repo
3	Experiment Designer	Creates control/variant setups, calculates sample sizes, designs test plans	VWO, Optimizely, AB Tasty
4	Experiment Monitor	Real-time health checks: SRM detection, velocity tracking, anomaly alerts	VWO MCP, Optimizely MCP
5	Analysis Engine	Statistical analysis, segment breakdowns, revenue impact calculations	VWO, GA4, data warehouse
6	Report Generator	Executive-ready reports with visualizations, decisions, and next steps	DOCX/PPTX/PDF output
7	Learning Repository	Searchable knowledge base of all experiments, patterns, and insights	Airtable, Notion, Obsidian
8	Next Best Test Engine	Auto-suggests new experiment IDs based on learnings, frameworks, and strategy	Internal logic + repository
9	xOS Dashboard	Real-time KPI/OKR tracking, experiment velocity, win rates, program health	Web frontend + all MCPs

4.1 Strategic Alignment Engine

This is the foundation of xOS and what makes it fundamentally different from any CRO tool on the market. Every experiment must trace back to a business objective. The Strategic Alignment Engine maintains a live map of:

- **Company OKRs/OPRs** — What the business is trying to achieve this quarter/year
- **KPI Tree** — How metrics cascade from top-line revenue down to micro-conversions
- **Opportunity Solution Tree** — Which opportunities map to which solutions (experiments)
- **Experiment-to-Goal Linkage** — Every test ID is tagged with the OKR and KPI it serves

When a new experiment is proposed, xOS automatically validates whether it serves the current strategic direction. When results come in, it shows exactly which business objectives were impacted. This eliminates “experimentation theater” — running tests that don’t move the needle.

4.2 Hypothesis Factory

The Hypothesis Factory takes a raw idea and transforms it through a structured pipeline:

Stage	Input	Process	Output
1. Capture	Raw idea, observation, or data signal	xOS asks clarifying questions to understand the intent	Structured idea card with context
2. Structure	Idea card	Applies hypothesis template: If [change], then [outcome], because [rationale]	Formal hypothesis with assumptions listed
3. Validate	Structured hypothesis	Checks against learning repository for similar past tests, validates strategic alignment	Validated hypothesis with risk flags
4. Prioritize	Validated hypothesis	Applies ICE/PIE/PXL scoring, factors in strategic weight and resource requirements	Prioritized and ranked in the experiment backlog
5. Assign	Prioritized hypothesis	Links to OKR, assigns unique test ID, creates experiment brief	Ready-to-execute experiment brief with ID

4.3 Experiment Designer

Once a hypothesis is approved, the Experiment Designer handles the technical experiment setup:

- **Control and variant configuration** — Defines what the control experience looks like and designs 1–N variants
- **Variant design suggestions** — Based on the hypothesis type (copy, layout, UX flow, pricing), xOS suggests specific variant designs drawing from the learning repository

- **Sample size and duration calculation** — Based on baseline metrics, minimum detectable effect, and traffic volume
- **Beta/staged rollout planning** — For high-risk experiments, designs a phased rollout: 5% > 25% > 50% > 100%

4.4 Experiment Monitor

Via MCP connections to experimentation platforms (VWO, Optimizely, AB Tasty), xOS monitors running experiments in real time:

- **SRM (Sample Ratio Mismatch) detection** — Alerts when traffic allocation is uneven, indicating a broken experiment
- **Velocity tracking** — Tracks whether conversions are accumulating at expected rates
- **Auto-stop recommendations** — Suggests stopping losing experiments early to save traffic for better tests

4.5 Analysis Engine

When an experiment concludes, the Analysis Engine does the full statistical workup:

- **Statistical significance testing** — Both Frequentist (p-value, confidence intervals) and Bayesian (probability to be best) approaches, following Tom Wessling's methodology
- **Segment analysis** — Breaks results down by device, geography, new vs returning, traffic source
- **Revenue impact projection** — Translates conversion lifts into annualized revenue impact with confidence ranges
- **Novelty and seasonal effect detection** — Flags when a "win" might be driven by novelty rather than genuine improvement

4.6 Report Generator

This is where xOS earns its keep with management. The Report Generator creates executive-ready documents that are so crisp and clear that decisions happen in the meeting, not two weeks later. Each report includes:

- **One-page executive summary** — The test, the result, the business impact, the recommendation. Done.
- **Visual data storytelling** — Charts and visualizations that tell the story without requiring a data science degree
- **Clear recommendations** — Ship it / Iterate / Kill it, with rationale
- **Next steps with new test IDs** — Automatically suggests follow-up experiments based on findings

Output formats: DOCX for stakeholder review, PPTX for presentations, PDF for archival. All generated automatically.

4.7 Learning Repository

Every experiment — win, lose, or inconclusive — gets cataloged in a structured repository (Airtable as the primary backend). The repository tracks the hypothesis, the setup, the

results, the learnings, and the follow-up actions. Over time, this becomes the company's experimentation memory — and it's what makes xOS exponentially more valuable the longer it runs.

Pattern recognition across experiments enables xOS to say things like: "In the last 12 months, 7 out of 9 tests that simplified the checkout flow resulted in a conversion lift of 5–15%. This pattern suggests high confidence in Hypothesis #127."

4.8 Next Best Test Engine

After every experiment concludes, xOS automatically generates new test ideas using multiple frameworks:

Framework	Logic	When to Use
Small Bets	Quick, low-risk iterations on proven winning patterns. Fast to implement, incremental gains.	When a test wins and there's room to optimize further (e.g., headline won test sub-headline)
Big Bets	Bold, strategic experiments targeting step-change improvements. Higher risk, higher reward.	When small bets plateau and the team needs breakthrough growth (e.g., entirely new checkout flow)
Compound Learning	Experiments designed specifically to generate learning, even if they don't win. Build understanding of user behavior.	Early-stage programs or new markets where the team lacks baseline knowledge
Follow-the-Data	Segment analysis reveals opportunities. If mobile lost but desktop won, design a mobile-specific test.	After any test with divergent segment performance

4.9 xOS Dashboard

A web-based dashboard that provides a real-time view of the entire experimentation program. Updated automatically via MCP connections. Key views:

- **Program Health Score** — Composite metric: velocity, win rate, revenue impact, strategic alignment
- **OKR/KPI Tracker** — How experiments are contributing to each business objective
- **Experiment Pipeline** — Backlog In Design Running Analyzing Decided
- **Revenue Impact Tracker** — Cumulative revenue impact from shipped experiments, broken down by quarter and OKR

5. The xOS Lifecycle: End-to-End Flow

Here is how a single experiment flows through xOS from start to finish:

Step	Phase	What xOS Does	Human Role
1	Ideation	Captures idea, asks clarifying questions, checks learning repository for prior related tests	Submits the idea (can be a one-liner)
2	Hypothesis	Structures into formal hypothesis, validates strategic alignment, flags risks	Reviews and approves hypothesis
3	Prioritization	Scores using ICE/PIE/PXL, ranks in backlog, suggests optimal test order	Final call on priority and sequencing
4	Design	Creates experiment brief: variants, sample size, duration, success metrics, guardrails	Reviews design, may adjust variants
5	Execution	Monitors experiment health via MCP: SRM checks, velocity tracking, anomaly detection	Implements the variant in the experimentation tool
6	Analysis	Full statistical analysis, segment breakdowns, revenue impact, significance testing	Reviews analysis for sanity check
7	Reporting	Generates executive-ready report with clear recommendation and next steps	Presents to stakeholders (or forwards the report)
8	Decision	Records decision, updates learning repository, triggers next-best-test suggestions	Makes the ship/iterate/kill decision
9	Next Cycle	Auto-generates 2–4 follow-up experiment IDs based on findings, updates backlog	Reviews suggestions, approves next batch

THE FLYWHEEL EFFECT

Every cycle makes xOS smarter. The learning repository grows. Pattern recognition improves. Recommendations get more accurate. After 6 months of running, a company's xOS brain knows more about their users' behavior than any individual CRO specialist ever could.

6. The xOS Marketplace

Beyond the core product, xOS opens up a massive opportunity: a marketplace where experimentation thought leaders sell their expertise as installable skills. Think of it as an App Store for experimentation intelligence.

6.1 How It Works

Each thought leader packages their unique methodology as an xOS skill — protected behind an MCP so the secret sauce stays secret. Companies can “rent” these expert brains and plug them into their xOS instance to enhance specific capabilities.

6.2 Launch Partners (Target)

Expert	Skill Domain	What Their Skill Does
Tom Wessling	Statistical Significance	Advanced statistical methodology for experiment analysis. His proprietary approach to when to call significance.
Erin Weigel	Experimentation Design	Best-in-class experiment design frameworks. How to design variants that actually test the hypothesis.
Lucas Vermeer	Strategic Direction	Strategic experimentation roadmapping. Connecting business strategy to experiment programs.
Bjarn Brunenberg	Strategic Setup in Orgs	How to set up and scale experimentation programs inside organizations. The operating model.

6.3 Revenue Model for Marketplace

Creators set their own pricing (monthly subscription or per-use). xOS takes a 30% platform commission. Creators earn passive income; companies get world-class expertise without hiring consultants. This also creates a network effect: more creators attract more companies, which attract more creators.

7. Technical Architecture (For Your Developer)

xOS is built entirely on Claude Code's skill and agent architecture. Your developer already knows this stack — here's how the pieces fit together.

7.1 Architecture Overview

Component	Implementation	Details
Core Brain	CLAUDE.md + Skills	Master system prompt with xOS personality, experimentation frameworks, and decision logic. This is the operating engine layer.
Module Skills	Claude Code Skills	Each of the 9 modules is a separate skill (SKILL.md + supporting scripts). Modular, testable, independently upgradeable.
Tool Integrations	MCPs	VWO MCP, Airtable MCP, Google Sheets MCP, GA4 MCP. Each integration is a separate MCP connection.
Autonomous Workflows	Agents	Multi-step workflows like "run full analysis" that chain multiple skills together. Agent orchestration for complex tasks.
Dashboard	Web Frontend	HTML/JS dashboard served locally or on a lightweight server. Pulls data via APIs from the learning repository (Airtable/Sheets).
Secret Sauce Protection	MCP Packaging	When productized, the operating engine is wrapped in an MCP. Clients get outputs, not source code. Critical for IP protection.

7.2 Skill Architecture (Folder Structure)

Recommended folder structure for the xOS repository:

Path	Purpose
CLAUDE.md	Master brain: xOS identity, core frameworks, agent routing logic
.claude/skills/strategic-alignment/	OKR/KPI tree management, opportunity solution tree, goal-experiment linkage
.claude/skills/hypothesis-factory/	Idea capture, hypothesis structuring, validation, prioritization

Path	Purpose
.claude/skills/experiment-designer/	Variant design, sample size calc, test plan creation
.claude/skills/experiment-monitor/	SRM detection, velocity tracking, health checks via VWO MCP
.claude/skills/analysis-engine/	Statistical analysis, segment breakdowns, revenue impact
.claude/skills/report-generator/	DOCX/PPTX/PDF report creation with data viz
.claude/skills/learning-repository/	Experiment catalog, pattern recognition, knowledge base
.claude/skills/next-best-test/	Auto-suggest new experiments, framework application
.claude/skills/dashboard/	Dashboard generation and update scripts
.claude/mcps/	MCP configurations for VWO, Airtable, Google Sheets, GA4
context/	Client-specific: OKRs, KPIs, brand guidelines, audience segments

7.3 Key Technical Decisions

- **Skills over agents for core modules.** Skills are deterministic, testable, and cacheable. Use agents only for complex multi-step orchestration that chains multiple skills.
- **Airtable as the primary data layer.** Structured enough for the learning repository, flexible enough for rapid iteration. MCP exists. Can migrate to a proper database later.
- **Immutable operating engine layer.** As Mateusz noted: core skills should have a stable API. Client context is mutable; the operating engine is versioned and controlled. This enables backward-compatible updates.
- **MCP wrapping for productization.** When selling to clients, the operating engine skills are exposed via MCP. Client gets the outputs (analyses, reports, suggestions) but not the underlying SKILL.md files or logic.

8. Development Roadmap (v1 in 8 Weeks)

Phased approach to get a working, sellable v1 within 1–2 months. Each phase builds on the previous one. The goal is to have something you can use on your own clients by end of Phase 3, and package for sale by end of Phase 4.

Phase 1: Foundation (Week 1–2)

Goal: Core architecture and the strategic alignment module.

Deliverable	Details
CLAUDE.md master brain	Core xOS personality, routing logic, experimentation frameworks encoded
Skill folder architecture	Full folder structure with placeholder SKILL.md files for all 9 modules
Strategic Alignment skill	OKR/KPI tree input, opportunity solution tree, experiment-goal linkage
Airtable MCP connection	Read/write experiments, OKRs, and hypotheses from Airtable

Phase 2: Hypothesis-to-Design (Week 3–4)

Goal: Complete pipeline from idea to ready-to-run experiment.

Deliverable	Details
Hypothesis Factory skill	Full pipeline: capture structure validate prioritize assign
Experiment Designer skill	Control/variant setup, sample size calculator, experiment brief generation
VWO MCP connection	Read experiment data, results, and configuration from VWO

Phase 3: Analysis + Reporting (Week 5–6)

Goal: From experiment completion to executive report in minutes.

Deliverable	Details
Analysis Engine skill	Statistical significance, segment analysis, revenue impact calculation
Report Generator skill	Executive report template (DOCX), data visualizations, recommendation engine

Deliverable	Details
Learning Repository skill	Experiment cataloging in Airtable, searchable knowledge base, pattern recognition

Phase 4: Intelligence + Polish (Week 7–8)

Goal: The flywheel features that make xOS uniquely valuable.

Deliverable	Details
Next Best Test Engine skill	Auto-suggest experiments using Small Bets/Big Bets/Compound Learning frameworks
Experiment Monitor skill	Real-time SRM detection, velocity tracking, anomaly alerts
xOS Dashboard v1	Web dashboard with program health, pipeline view, KPI tracking
End-to-end testing	Run full lifecycle on 2–3 real experiments. Fix edge cases. Polish flows.

9. Go-to-Market Strategy

9.1 Pricing Model

Tier	Price	Includes
Implementation	€15–25K one-time	Context setup (OKRs, KPIs, tool integrations), custom configuration, training the 2 human operators, first 5 experiments guided
Monthly License	€2–5K/month	xOS brain access, all core modules, MCP integrations, dashboard, updates
Marketplace Skills	Varies (set by creator)	Add-on expert skills (Tom Wessling stats, Erin design, etc.). 30% platform commission.
Enterprise	Custom	Multi-team deployment, custom integrations, dedicated support, SLA

9.2 Sales Motion

- **Prove it on your own clients first.** Use xOS on 2–3 freelance clients during Phase 3–4 development. Build case studies with real numbers.
- **LinkedIn thought leadership.** Document the build in public. Share experiment results, dashboard screenshots, time-saved metrics. Your community is already watching.
- **Pilot program (5 companies).** Offer discounted implementation (€10K) to 5 mid-market companies in exchange for feedback and case studies. Target companies with 3+ CROs.
- **Scale.** With 5 case studies showing cost reduction and velocity improvement, sell to the broader market. Launch marketplace with 3–4 thought leader skills.

9.3 The Pitch (One Sentence)

ELEVATOR PITCH

xOS is an AI experimentation brain that replaces 80% of your CRO team's manual work – from strategic planning through test design, analysis, and executive reporting – so you get faster decisions, compounding learnings, and 3x experiment velocity with 2 operators instead of 10.

10. Why xOS Wins

Competitor/Alternative	What They Do	What xOS Does Differently
VWO / Optimizely	Run experiments. That's it. No strategy, no analysis, no reporting, no learning.	xOS uses these as data sources via MCP. It's the brain on top of the tools.
In-house CRO team	Manual everything. Expensive. Knowledge lives in people's heads. Turnover kills the program.	xOS captures all knowledge permanently. No turnover risk. 80% cost reduction.
CRO agencies	Outsourced expertise. Slow. Expensive. They leave and take the knowledge.	xOS is an in-house brain that gets smarter over time. No dependency on external agencies.
Generic AI (ChatGPT/Claude)	Can help with individual tasks but has no experimentation expertise, no tool integration, no memory across experiments.	xOS is purpose-built for experimentation. Specialized skills, tool integrations, compounding knowledge.

11. Immediate Next Steps

- **Share this document with your developer.** Section 7 (Technical Architecture) and Section 8 (Roadmap) are their starting points. Phase 1 starts immediately.
- **Set up the GitHub repository with the folder structure from Section 7.2.** Placeholder SKILL.md files for all 9 modules. Get the architecture right from day one.
- **Write the CLAUDE.md master brain.** This is your brain dump: experimentation philosophy, frameworks, decision logic. This is what makes xOS think like Bjarn.
- **Connect Airtable MCP.** This is the data backbone. Get it working early so every skill can read/write experiments.
- **Reach out to Mateusz about the Second Brain Factory collaboration.** He offered to let you test-drive their packaging tool. Your xOS could be one of the first brains on their marketplace.

xOS – The Experimentation Operating System

From chaos to system. From 10 specialists to 2 pilots. From gut feeling to compounding intelligence.